Validation and AI Use

Known-answer benchmark, accounting and valuation checks, locked prediction, AI-output check, partner challenge, and reproducibility record

This is the frozen-format validation artifact assembled from current records. Incomplete evidence remains explicitly incomplete; most importantly, the assignment-required recorded fresh-environment run has not been supplied.

Target	The Home Depot, Inc. (NYSE: HD)
Decision user	Buy-side investment committee; no current position
Valuation date	September 1, 2026
Currency / share basis	USD millions; estimated valuation-date diluted shares
Current decision	WATCH / DEFER
Model version checked	Repository commit 0a4c113215fdda3efdf075926de03e919e7e89e4

Educational AI finance class work; not professional financial advice.

Decision, version, and evidence boundary

Field	Record
Decision	Watch/defer; do not initiate at the September 1, 2026 close of $319.77
Operating DCF range	$220.99-$253.91 per diluted share; base $234.00
Comparable references	$240.00 P/E; $217.21 EV/EBITDA; not averaged with DCF
Valuation boundary	September 1, 2026; evidence public by that date
Model basis	USD millions; estimated valuation-date diluted shares; FCFF/WACC
Code commit checked	0a4c113215fdda3efdf075926de03e919e7e89e4
Official memo	Project_1_Decision_Memo_HD.pdf; SHA-256 226c54e570b6a3e816b59157043c352cea8422502f0de3ac5c19af50aa1956d7

Published training-case benchmark

Source: course-materials/lessons/week-03/starter/dcf_case.csv and dcf_expected_output.csv. Convention: USD millions, year-end FCFF, five explicit years, Gordon terminal value at Year 5.

Output	Actual	Expected	Result
FCFF Years 1-5	108.000000; 114.480000; 120.204000; 125.012160; 128.762525	Same	PASS
PV explicit FCFF	448.440817	448.440817	PASS
Terminal value at Year 5	1,894.648579	1,894.648579	PASS
PV terminal value	1,176.427703	1,176.427703	PASS
Enterprise value	1,624.868520	1,624.868520	PASS
Common equity value	1,374.868520	1,374.868520	PASS
Value per diluted share	27.497370	27.497370	PASS
Terminal-value share	0.724014	0.724014	PASS

Overall: PASS. Evidence: benchmark_validation.py and 14_training_benchmark_output.txt.

Accounting, valuation, and failure checks

Claim / failure mode	Test	Result	Evidence
Reported FY2025 base	Individual income-statement and balance-sheet component sums; CFO + CFI + CFF and cash rollforward	PASS; all gaps 0	home_depot_fcff_model.py; 02_implementation_results.md
Forecast articulation	Balance-sheet gap and cash-flow-to-cash gap in every year/case	PASS; all gaps 0	Model; 05_part3_visible_output.txt
Signed FCFF	Synthetic -$100m first-year FCFF must reduce EV rather than be floored	PASS	Core validation output
Debt bridge	$1,000m more debt must reduce equity value $1,000m; finance leases included once	PASS	Model; 03_valuation_basis.md
DCF direction	Value falls as WACC rises and rises as terminal growth rises	PASS	Part 3 sensitivity
Terminal boundary	Reject terminal growth greater than or equal to WACC and exit unsuccessfully	PASS	Core/Part 3 boundary test
Case determinism	Restore/rerun base after scenarios	PASS	part3_analysis.py
WACC implementation	Calculated WACC must equal BASE_INPUTS['wacc']	PASS; 8.1984%	home_depot_fcff_model.py

Highest-risk claim and failure test

The highest-risk valuation claim is the terminal-value dependence: 76.57% of base enterprise value occurs after 2030. The engine refuses g >= WACC rather than producing an invalid Gordon value. The separate locked changed-input test also measures the economic impact of reducing terminal growth. Both tests pass; neither resolves whether 2.5% is the correct economic assumption.

Locked Changed-Input Record

Pre-run record - preserved before execution

Field	Record
Recorded at	October 19, 2026, 18:18:31 EDT
Precommit	46e3d779148b70b78a9ea9bde56ff9a9c7674c03; committed 18:18:48 EDT
Input	Terminal growth: 2.5% to 1.5% per year (-1.0 percentage point)
Held fixed	Five-year operating assumptions/FCFF, WACC, excess cash, debt, diluted shares, and discount timing
Original student prediction	I expect value per share to fall a little like around 3%.

Actual result and decision effect

Measure	Base	Changed	Change
Value per diluted share	$234.00	$199.38	-$34.62 / -14.80%
Enterprise value	$286,819.6m	$252,209.8m	-$34,609.8m
Terminal-value share	76.57%	73.36%	-3.22 points

The predicted direction was correct, but the decline was about 11.80 percentage points larger than the approximate 3% prediction. The watch/defer action did not change because both values remain below $319.77. The mismatch increases the priority of terminal-growth and WACC research. The initial failed command used the wrong relative virtual-environment path; the rerun corrected the execution path without changing the model or input.

Checked AI output

Field	Record
Tool/model surface	Codex, GPT-5; no finer model version exposed
Interaction date	October 19, 2026
Material AI output	Tractor Supply carried forward as a potential Home Depot comparable
Independent evidence	Tractor Supply Q2 FY2026 10-Q, Note 8 product-category table: 31% livestock/equine/agriculture; 23% seasonal/recreation; 23% companion animal; 15% truck/tool/hardware; 8% clothing/gift/decor
Source	https://www.sec.gov/Archives/edgar/data/916365/000091636526000059/tsco-20260627.htm
Disposition	Qualified, not accepted as a close peer
Decision effect	Peer median remains a fragile reference; no mechanical average; watch/defer unchanged

This example checks peer classification only. It is not a check of the market-price source or the entire comparable valuation. The debt correction is documented as model work elsewhere, but is not relabelled as the recorded AI-output example.

Credited partner feedback

Field	Record
Partner/date	Jack Schlueter; October 13, 2026
Student-reported challenge	The biggest valuation risk was the 76.57% terminal-value share; recorded as a paraphrase, not a verbatim transcript
Student response	His point is correct that the valuation leans heavily on the terminal value (76.57%). Your own test supports it, because you predicted a drop of about 3% and got 14.80%.
Independent check	76.57% terminal share and -14.80% locked-test result match the model/record
Disposition/effect	Accepted; recommendation unchanged; terminal growth and WACC elevated as research priorities

Provenance qualification: this interaction is documented from the student's recollection. No independent timestamp or transcript was supplied, and Edition A is not edited retroactively.

Reproducibility and cold-run record

Item	Status
Repository	https://github.com/Ethan-Todd/Project-1-HD
Environment	Standard-library model; requirements.txt declares no project packages
Unrecorded clean clone	A clean clone of published commit 192d040... contained required files; benchmark, core model, and Part 3 all exited 0; later reviewed model scripts retain the same content hashes
Benchmark	Twelve PASS results; Overall PASS
Core model	Historical, balance, cash, signed-FCFF, debt, monotonicity, bridge and lease checks PASS
Part 3	All cases balance; sensitivity, deterministic rerun, debt, boundary and negative-FCFF checks PASS
Required recorded run	INCOMPLETE - no actual recording or recording URL supplied
Submission placement	UNVERIFIED - checklist has no fourth-video row; Video 2 requires reproducibility/high-risk validation but does not explicitly place the full run there

Exact run sequence

```
git clone --branch main --single-branch https://github.com/Ethan-Todd/Project-1-HD.git Project-1-HD
cd Project-1-HD
git rev-parse HEAD
python3 -m venv .venv
./.venv/bin/python -m pip install -r requirements.txt
./.venv/bin/python benchmark_validation.py
./.venv/bin/python home_depot_fcff_model.py
./.venv/bin/python part3_analysis.py
git status --short
```

AI assistance disclosure

Codex (GPT-5 surface; no finer version exposed) assisted after the Edition A boundary with source organization, code/model implementation, debugging, validation design and execution, document drafting, repository operations, consistency review, and PDF assembly. The student supplied the human-authored locked prediction and the partner recollection and owns the forecast judgments, peer policy, recommendation, and all submitted claims. AI outputs were not accepted merely because they sounded technical; the recorded TSCO peer suggestion was checked against an SEC filing and qualified.

Material limitations and missing evidence

- Recorded clean end-to-end fresh-environment run and its submission/recording reference.
- Videos 1-3, corrected transcripts, durations, timestamp indices, and logged-out access checks.
- Verified Edition A Brightspace receipt or contemporaneous submission evidence; matching hashes do not prove originality.
- Independent timestamp/transcript for Jack Schlueter's feedback, if required.

- Exchange-quality September 1 prices; current amounts are cross-verified using third-party tables.
- Frozen capture of the official dated ERP statement and reconciliation of its 4.75% Treasury input with the model's independent 4.79% par-yield convention.
- Resolved beta, marginal debt cost/tax, exact award dilution, excess cash, and operating-lease rollforward.
- Final repository commit/URL fields, video IDs, and complete logged-out access confirmation.

Monitoring and reversal

Monitor comparable sales and transactions, big-ticket/project demand, gross margin, SG&A; leverage, acquisition integration/contribution, inventory days, capex, and market-based WACC inputs. Initiate only if a refreshed valuation using sourced results and resolved WACC shows sufficient upside to the then-current price. Move toward do-not-initiate if downside demand/margin pressure is met or exceeded, acquisition integration weakens, or a material accounting/bridge validation fails.

Primary evidence locations

04_part3_assumption_challenge_record.md; 05_part3_visible_output.txt; 07_comparable_valuation.md; 08_locked_changed_input_record.md; 11_ai_output_check.md; 13_source_consistency_audit.md; 14_training_benchmark_output.txt; 20_partner_challenge_status.md; 21_edition_a_provenance.md; 22_decision_memo_verification.md; benchmark_validation.py; home_depot_fcff_model.py; part3_analysis.py.

Source and definition notes

- Home Depot FY2025 Form 10-K: year ended February 1, 2026; filed March 18, 2026. Q2 FY2026 Form 10-Q: balance sheet August 2, shares August 18, filed August 25, 2026.
- Financing debt includes commercial paper, current and noncurrent long-term debt, and finance leases once. Operating leases remain excluded consistently because lease costs remain in operating profit and FCFF.
- The 4.79% risk-free rate is the September 1 U.S. Treasury 10-year par yield. The 4.14% ERP is Damodaran's dated implied estimate; beta and marginal debt/tax remain proxies.
- September 1 closing prices are cross-verified using third-party historical tables. The lack of an exchange-quality saved archive remains disclosed rather than treated as resolved.

Cross-artifact consistency

The official memo, visible output, model scripts, locked record, and these PDFs use the same September 1, 2026 valuation date, $319.77 market price, $220.99-$253.91 operating range, $234.00 base, $240.00 P/E reference, $217.21 EV/EBITDA reference, 8.1984% provisional WACC, 2.5% terminal growth, 76.57% terminal-value share, +6.23-point reverse-DCF shift, and watch/defer recommendation.